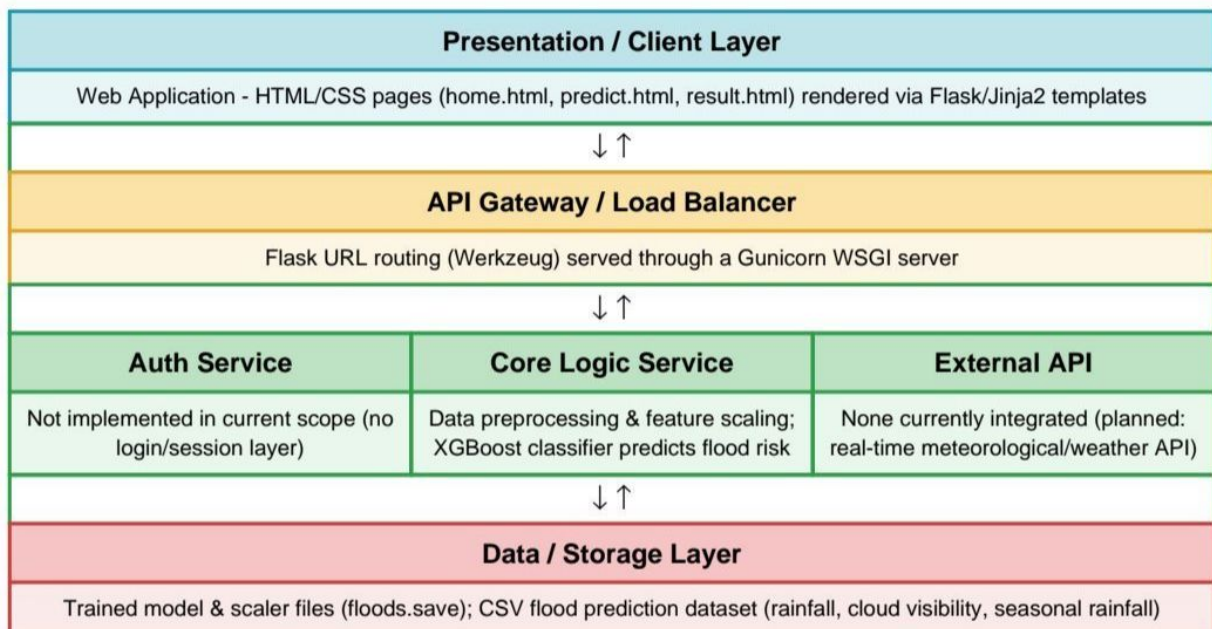


Solution Architecture

Date	01 July 2026
Team ID	xxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Renders the user interface for entering rainfall/weather details and displaying the flood risk prediction result.	HTML, CSS, Flask (Jinja2 templates)
<i>API Gateway</i>	Routes incoming HTTP requests between the client and the core prediction logic; serves the application.	Flask routing, Gunicorn (WSGI server)
<i>Microservices</i>	Preprocesses rainfall/weather input, scales features, and runs the trained ML model to predict flood risk.	Python, XGBoost, scikit-learn, NumPy, Pandas, Joblib/Pickle
<i>Database</i>	Stores the historical rainfall/weather data used to train the model, and the serialized model/scaler files.	CSV flood prediction dataset (Kaggle), local .save/.pkl file storage